

CryptoLabX — Cryptanalysis of Shift Cipher

Course: Cryptography Laboratory (22CPP307) | Assignment 4 Report

1. Purpose of Laboratory Exercise

The objective of this laboratory exercise is to perform cryptanalysis on the Shift Cipher (Caesar Cipher) using two distinct automated recovery techniques: **Exhaustive Brute Force with Dictionary Scoring** and **Chi-Square (χ^2) Letter Frequency Goodness-of-Fit Analysis**. The lab evaluates the efficacy, limits, and failure modes of both techniques across varying text lengths and structural characteristics.

2. Cryptanalysis Algorithms

A. Brute-Force & Dictionary Scoring Attack

The Shift Cipher has a key space of size $|K| = 26$. The brute-force algorithm tests every key $k \in [0..25]$. For each key, the ciphertext is decrypted: $P_candidate = (C - k) \bmod 26$. The decrypted text is tokenized into words and compared against an English dictionary (*english_words.txt*). The key yielding the highest count of valid English words is selected as the predicted key.

B. Chi-Square (χ^2) Letter Frequency Analysis

Chi-Square analysis evaluates how closely letter frequencies of candidate decrypted text match expected English monogram probabilities (e.g., E $\approx$ 12.7%, T $\approx$ 9.05%). The formula used is:

$$\chi^2 = \sum [(\text{Observed}_i - \text{Expected}_i)^2 / \text{Expected}_i] \text{ for } i = \text{'A' to 'Z'}$$

Where $\text{Expected}_i = N \times \text{Probability}_i$ and N is the total count of alphabetic characters in the candidate. The shift key k that produces the **minimum χ^2 statistic** is chosen as the predicted plaintext key.

3. Experimental Results Table

Test Case #	Actual Key	Dictionary Key	Chi-Square Key	Dict Correct?	Chi-Square Correct?
1 (Medium Text)	7	7	7	YES	YES
2 (Long Technical)	15	15	15	YES	YES
3 (Short Text)	3	3	3	YES	YES
4 (No Spaces)	12	0	12	NO	YES
5 (High Z/J Text)	19	19	25	YES	NO

4. Comparison of Cryptanalysis Methods

- Text Without Word Boundaries (Spaces):** Chi-Square successfully recovered Key 12 in Test Case 4 where Dictionary Scoring failed. Dictionary scoring relies on whitespace tokenization, while Chi-Square acts directly on character distributions.

- **Unusual / Non-Standard Frequency Texts:** Dictionary Scoring correctly predicted Key 19 in Test Case 5 where Chi-Square failed. Texts with heavy occurrences of rare letters ('Z', 'J', 'Q') distort letter frequency statistics.

5. Failure Analysis & Recommended Improvements

Why Dictionary Attack Failed (Test Case 4): Without word spaces, tokenizer treats the entire text as a single word.

Improvement: Implement sliding-window string segmentation or n-gram frequency scoring.

Why Chi-Square Attack Failed (Test Case 5): Short texts featuring atypical letter distributions produce a high χ^2 score for the true plaintext. *Improvement:* Combine Bigram/Trigram frequency matrices with dictionary scoring for hybrid verification.

6. Observations & Conclusion

Observations: Chi-Square requires at least 30-50 characters for statistical significance. Dictionary attack works well for short texts with standard word spacing.

Conclusion: Shift Cipher provides zero security against automated cryptanalysis due to its minimal key space ($|K|=26$) and preserved structural letter frequency.